

Basepoint independence without cohomology, and why $c_f(0) = 0$ is a subtraction, not a restriction

The one identity that does the work

Define the linear functional on a two-segment finite contour,

$$L^*(f, s; \tau_0) = \int_{C_S(\tau_0)} f(\tau) \tau^{s-1} d\tau + \int_{C_T(\tau_0)} f(\tau) \tilde{k}_T(\tau, s) d\tau,$$

with the Hurwitz T -kernel built from two terms, $\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - \varpi_k \zeta(s-(k-1), \tau+1)$, where ϖ_k is the constant S -fold weight phase ($\varpi_k = i^k$ in the convention of arXiv:1806.09874; *not* an s -dependent factor). The kernel exists to satisfy a single difference equation under $\delta_T g(\tau) := g(\tau+1) - g(\tau)$:

$$\delta_T \tilde{k}_T(\tau, s) = -(\tau+1)^{s-1} + \varpi_k (\tau+1)^{k-1-s}.$$

The two terms of $\tilde{k}_T$ are present precisely to produce *both* powers, τ^{s-1} and τ^{k-1-s} — enough to cancel both the direct and the S -image boundary terms below. (Exact prefactors are contour-specific; read them off the actual geometry.)

δ_T -inversion *is* basepoint independence

These are not two facts. Vary the base point and differentiate. The T -segment endpoints move together, and the only property of f used is T -periodicity $f(\tau_0+1) = f(\tau_0)$, so the T -boundary variation is

$$\partial_{\tau_0} \int_{C_T} f \tilde{k}_T = f(\tau_0) \delta_T \tilde{k}_T(\tau_0, s).$$

This is engineered to cancel the S -segment variation mode by mode. Every Fourier mode of f cancels except the constant, leaving

$$\boxed{\partial_{\tau_0} L^*(f, s; \tau_0) = c_f(0) \Phi(\tau_0, s)}$$

with Φ *universal* — independent of f , of its principal part, of everything but the single coefficient $c_f(0)$. So “ $\tilde{k}_T$ inverts δ_T ” and “ L^* is τ_0 -independent” are the same statement. The cohomology was only the name; the difference equation is the content, and it survives stripping the cohomology away.

$c_f(0) = 0$ is a subtraction, not a restriction

The boxed identity says the entire base-point dependence is the one constant mode. Hence, by linearity,

$$L^*(f - c_f(0), s; \tau_0) = L^*(f, s; \tau_0) - c_f(0) L^*(1, s; \tau_0)$$

has $\partial_{\tau_0} = c_f(0)\Phi - c_f(0)\Phi = 0$. Therefore **every** meromorphic modular form — any pole structure, constant term present or not — has a base-point-independent L^* once the constant mode is removed. The subtracted piece $c_f(0) L^*(1, s; \tau_0)$ is explicit, and is exactly where the poles of Λ sit. Nothing is discarded; the constant is relocated to the polar part.

Precision (do not skip when writing the proof). $f - c_f(0)$ is *not* a modular form: subtracting a weight-0 constant from a weight- k form breaks the S -law. So the independence is *not* justified by calling $f - c_f(0)$ “the cusp form.” It is justified by linearity together with the universal-mode identity above — i.e. by δ_T -inversion. What the cancellation actually uses, and what *is* preserved under the subtraction, is T -periodicity. Keep those two straight: T -periodicity survives the subtraction and is used; S -modularity is broken by it and is not what carries the argument.

The poles come from the kernel, not from a divergence

Classically, with $\Lambda(f, s) = \int_0^\infty (f(iy) - c_0) y^{s-1} dy$, one splits at $y = 1$ and folds the lower piece by $f(i/y) = (iy)^k f(iy)$; the $(f - c_0)$ integrals are entire and the leftover elementary integrals of the constant give

$$\Lambda(f, s) = (\text{entire}) - \frac{c_0}{s} - \frac{i^k c_0}{k - s},$$

a simple pole at $s = 0$ with residue $-c_0$ and one at $s = k$ with residue $i^k c_0$. These sit on top of the $n \neq 0$ incomplete-gamma sum without touching it: the BFK part is entire, the poles are purely the $n = 0$ mode.

In the finite contour there is no cusp, so the poles cannot come from a divergence at $y \rightarrow 0, \infty$. They come from the kernel. The Hurwitz function $\zeta(\sigma, a)$ has its only pole at $\sigma = 1$, with residue 1 independent of a . The first kernel term $\zeta(1 - s, \tau + 1)$ meets it at $s = 0$; the second Hurwitz term, carrying the S -fold weight phase, meets it at $s = k$. Near $s = 0$,

$$\int_{C_T} f(\tau) \tilde{k}_T(\tau, s) d\tau \sim -\frac{1}{s} \int_{\tau_0}^{\tau_0+1} f(\tau) d\tau,$$

and the unit-length T -segment annihilates every nonzero Fourier mode, $\int_{\tau_0}^{\tau_0+1} e^{2\pi i n \tau} d\tau = 0$ for $n \neq 0$, leaving exactly c_0 . So the residue at $s = 0$ is $-c_0$, and it is *basepoint-independent* even though the regular part of $c_0 L^*(1; \tau_0)$ is not. The $s = k$ pole arises the same way through the second term; its residue is $i^k c_0$.

This matches McGady, arXiv:1806.09874, Thm. 1.2:

$$L_f^*(s)^{\text{reg}} = -c_f(0) \left(\frac{B^s}{s} + \frac{i^k B^{k-s}}{k-s} \right) + \text{regular}(s) = i^k L_f^*(k-s)^{\text{reg}},$$

so residue $-c_f(0)$ at $s = 0$, residue $+i^k c_f(0)$ at $s = k$, and functional equation $L^*(s) = i^k L^*(k-s)$, for every weight- k meromorphic form regular at cusps.

Origin of the i^k . The phase is the S -fold factor, weight-indexed, not s -indexed. On the imaginary axis $\tau = iy$, modularity $f(-1/\tau) = \tau^k f(\tau)$ folds the contour with $S(i/y) = iy$ and pulls out $(1/i)^k = (-i)^k$ — a constant in s depending only on the weight. That is why it survives as i^k for *all* even k and attaches to the $s = k$ pole rather than the $s = 0$ one. Any s -dependent phase of the form $e^{i\pi(s-1)}$ is wrong here: at integer s it is ± 1 and can never produce a factor of i . The second kernel term’s prefactor must be the constant S -fold weight phase; take its exact form from the fold, not from an s -dependent reconstruction.

Summary. The variation-in- T operator is load-bearing, not an artifact. It is one operator seen on two slices: on V_{k-2} for the period polynomial (integer s), and on functions with Hurwitz ζ as the antidifference of τ^{s-1} for the L -function (all s). Its single difference equation is simultaneously the parabolic gauge, the base-point independence, and the reason a constant-term subtraction — rather than a hypothesis on f — suffices for arbitrary pole structure.